

Estuary Sensor Drift Study

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Abstract

Dissolved oxygen records can inherit temperature-dependent drift during long estuary deployments. We compared three optical stations under matched tidal cycles and tested a correction that retains the observed diel signal without overstating precision.

1. Study design

Stations E-01, E-02, and E-03 sampled every 15 min for 21 days. Calibration water was prepared at 20 °C, and all probes were exchanged between stations after day 7 [1].

1.1 Drift definition

The drift rate was estimated only after stable salinity and pressure windows were confirmed. Negative excursions during maintenance were excluded before the slope was fitted.

1.2 Quality control

Replicate bottles constrained the sensor comparison. A station was retained when the paired difference remained below 0.18 mg L⁻¹ for at least 90 % of valid intervals.

Interpretive limit

The correction reduces systematic offset; it does not convert field observations into an independent reference measurement.

$$R_t = (C_t - C_0) / \Delta t$$

$$D_{corr} = D_{raw} - \beta(T - 20)$$

Footnote 1. Clock offsets were checked against a local reference before deployment.

2. Results

Median uncorrected drift differed among stations, but its sign was stable within each instrument. E-02 showed the largest temperature response and the narrowest residual distribution after correction.

2.1 Tidal phase

Oxygen minima followed the flood-to-ebb transition by 42 min (interquartile range 31-56 min). This lag remained after correction and was therefore treated as environmental rather than instrumental.

Station	Raw	Corrected	n
E-01	+0.09	+0.02	1911
E-02	+0.16	+0.03	1898
E-03	-0.07	-0.01	1904

Table 1. Drift estimates in $\text{mg L}^{-1} \text{ day}^{-1}$. Signs and decimal precision are retained from the instrument export.

2.3 Sensitivity

Changing the stability window from 30 to 45 min altered β by less than 0.004. Removing the highest-salinity day did not reverse any station ranking.

2.2 Residual checks

Residuals were centered within the prespecified tolerance at all three stations. The 95 % intervals widened during the warm afternoon period, especially when current speed was low.

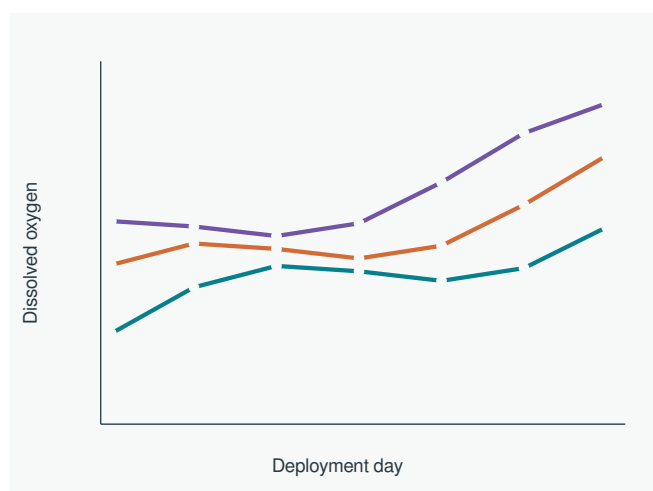


Figure 1. Station-specific dissolved oxygen trajectories after temperature correction. Lines show medians; symbols identify daily checks.

3. Discussion

The paired deployment separates sensor drift from much of the tidal structure, but the experiment does not isolate biofouling from optical aging. The correction should therefore be transferred only with a fresh bottle check.

3.1 Scientific interpretation

A small corrected drift does not imply that all short oxygen excursions are biological. Flow shading, probe orientation, and transient bubbles remain plausible causes [2].

Uncertainty

Uncertainty combines the slope standard error and bottle-reference variance. It excludes uncertainty from spatial gradients between the pier and channel stations.

3.2 Operational guidance

Field teams should store the raw signal, the correction coefficient, and calibration temperature together. Reprocessing must never overwrite the original observation series.

Limitations

The 21-day study spans one neap-spring cycle and cannot establish seasonal behavior. The tested salinity range was 12.4-27.8 PSU.

Data availability

Synthetic records and the processing specification are identified as dataset EST-DRIFT-26. The identifier is descriptive and does not resolve to a public archive.

4. Conclusions

Temperature correction reduced station-specific dissolved oxygen drift while preserving the repeated tidal phase relationship. The result supports transparent reprocessing, not replacement of field calibration or independent quality checks.

References

- [1] Lin R, Perez J. Optical oxygen checks under variable salinity. *Coastal Methods* 12:44-52. 10.5555/estuary.2026.014.
- [2] Okafor M. Interpreting short dissolved oxygen excursions. *Estuarine Signals* 8:101-109.
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Author contributions

M.V. designed the field comparison; O.P. verified calibration records; both authors reviewed the final synthetic report.